# Cool summers are stormy summers: precession reorganizes extratropical cyclone activity through the quiet season


## Abstract

Extratropical cyclones peak in winter, while orbital precession reshapes the seasonal insolation cycle most strongly in summer. Thus storm tracks might be expected to ignore the precession cycle. This is examined with twelve equilibrium simulations using a coupled climate model AWI-ESM2, spanning a full precession cycle at 30° longitude increments under fixed eccentricity and obliquity. Extratropical cyclones are explicitly tracked based on 6-hourly atmospheric pressure fields for both hemispheres. While the annual-mean cyclone response to precession is muted, substantial seasonal variability emerges. In the North Atlantic, the North Pacific, and the Southern Ocean the response concentrates in local summer, where storm activity ranges across the phases by up to 68% of its climatological mean, while winter changes stay below about 10%. We also find summers spent near perihelion feature suppressed storminess, whereas summers near aphelion experience intensified cyclone activity, yielding antiphase midlatitude storm variability between the two hemispheres over the precession cycle. This signal is carried by cyclone number and position rather than by propagation speed, and it follows the pronounced summer response of the Eady growth rate driven primarily by vertical shear in all three basins. Monthly storm responses are phase-locked to the calendar timing of perihelion, lagged by 1–2 months due to surface thermal inertia. Precession therefore acts on extratropical storm tracks through their quiet season. Summer-sensitive proxy archives should record a strong and hemispherically antiphased storminess signal at the precession period that annually integrating archives would miss.

# 1 Introduction

Extratropical cyclones organize the weather of the middle latitudes. They carry most of the poleward heat and moisture transport outside the tropics, deliver a large share of midlatitude precipitation, and cluster into the storm tracks of the North Atlantic, the North Pacific, and the Southern Ocean [Hoskins and Valdes, 1990, Chang et al., 2002, Hoskins and Hodges, 2002, 2005]. These storms grow by baroclinic instability on the meridional temperature gradient [Eady, 1949], and the gradient is steepest in winter. Cyclone numbers therefore peak in the cold season in both hemispheres, in the south as clearly as in the north [Simmonds and Keay, 2000]. Extratropical cyclones are creatures of winter.

Precession, in contrast, is a lever that acts on summer. The slow turning of the longitude of perihelion is the strongest orbital pacemaker of the seasonal insolation cycle, yet it redistributes sunlight rather than adding any, and the redistribution is proportional to the sunlight a season already receives [Berger, 1978, Laskar et al., 2004]. Midlatitude summer insolation swings by more than 100 W $m^{-2}$ across a precession cycle while midwinter insolation barely moves, which is why the orbital literature treats summer energy as the climatically active quantity [Huybers, 2006]. A winter phenomenon facing a summer lever invites a simple expectation. Precession should have little to say about storm tracks. That expectation has never been put to a clean test.

Where the imprint of precession has been studied in depth, it appears as a seasonal reorganization of climate rather than a change in the annual mean. The monsoon literature carries this tradition, from the classic orbital experiments of Kutzbach and Guetter [1986] to single-forcing designs that isolate precession explicitly [Bosmans et al., 2015, Merlis et al., 2013], and cave records show the northern and southern monsoons swinging in antiphase as perihelion migrates through the year [Wang et al., 2008]. The extratropics have received far less attention. Kaspar et al. [2007] simulated a stronger and poleward-shifted North Atlantic winter storm track for the Eemian from two orbital time slices, Brayshaw et al. [2010] traced Holocene orbital forcing into the North Atlantic and Mediterranean winter storm tracks through the baroclinicity they support, and Varma et al. [2012] found the Southern Hemisphere westerlies migrating under Holocene orbital forcing. Explicit cyclone tracking has entered paleoclimate work mainly at the Last Glacial Maximum, where ice sheets dominate the forcing [Pinto and Ludwig, 2020, Raible et al., 2021], and in transient simulations of the last millennium [Hess and Chemke, 2025]. For tropical cyclones the precessional response has recently been isolated in high-resolution paleoclimate simulations [Raavi et al., 2023]. To our knowledge, no previous study has applied explicit cyclone tracking across a full precession cycle to isolate the orbital response of the extratropical storm tracks.

Closing this gap matters beyond dynamics. However, the paleoclimate records that would register past storminess, from dust and wind-blown sediment to westerly-belt precipitation, respond to particular seasons rather than to the annual mean, and a seasonally biased archive can fabricate or erase an orbital signal depending on which season carries it [Laepple et al., 2011]. Knowing where in the seasonal cycle the precessional storm signal lives is a prerequisite for reading such records. Storm diagnostics raise a parallel issue, because measures of cyclone influence time, cyclone counts, and cyclone intensity respond to forcing in different ways and must be separated before any single proxy is interpreted.

In the present study the precession cycle is covered with twelve equilibrium simulations of a coupled climate model, and every extratropical cyclone is tracked explicitly in six-hourly pressure fields in both hemispheres. The design isolates precession at amplified eccentricity with all other boundary conditions fixed, so that the seasonal anatomy of the orbital storm response can be mapped basin by basin against the calendar month of perihelion.

The naive expectation fails, and it fails in an instructive way. The annual-mean storm response is indeed small, but the signal has not vanished. It is concentrated in local summer, the quiet season, in all three storm basins. Summers spent near perihelion are calm and summers spent near aphelion are stormy, so the two hemispheres respond in antiphase across the precession cycle, an extratropical counterpart of the monsoon seesaw. The response travels from insolation to storms through the summer Eady growth rate, and the growth-rate change is carried by vertical shear [Lindzen and Farrell, 1980]. The paper is organized around which season carries the precessional storm response, what dynamical chain transmits it, and why the winter storm track stays orbitally quiet. Section 2 describes the model, the ensemble, and the tracking. Section 3 presents the seasonal anatomy of the response, and section 4 discusses mechanisms, implications for proxy records, and limitations.

# 2 Data and methods

## 2.1 Model and experiments

The simulations are performed with the coupled Earth system model AWI-ESM2 [Sidorenko et al., 2019, Shi et al., 2022]. The atmosphere component is ECHAM6 at T63 resolution with 47 vertical levels [Stevens et al., 2013], which includes the land surface module JSBACH [Raddatz et al., 2007]. The ocean and sea ice component is FESOM2, a finite-volume model formulated on an unstructured mesh [Danilov et al., 2017].

Twelve equilibrium simulations are analyzed, taken from the idealized precession ensemble introduced by Shi et al. [2025]. The longitude of perihelion $\omega$ is prescribed at values from 0° to 330° in steps of 30°, while eccentricity is fixed at 0.058 and obliquity at 24.5°. The eccentricity value sits near the upper bound reached during the Quaternary and amplifies the precessional modulation of insolation, so that the response emerges clearly from internal variability. All remaining boundary conditions follow the preindustrial setup, and each phase is branched from a long preindustrial control. Thirty years of six-hourly output from the equilibrated part of each simulation form the basis of the cyclone analysis. The calendar month of perihelion for each phase is computed from two-body orbital geometry [Berger, 1978], and the same formulary provides the insolation fields shown in Fig. 1.

## 2.2 Cyclone detection and tracking

Extratropical cyclones are detected and tracked with the algorithm of Crawford et al. [2021], and Fig. 2 summarizes the procedure with one six-hourly field of the ensemble. Six-hourly sea level pressure is first interpolated onto 100-km equal-area grids centered on each pole, and the two hemispheres are processed separately. A grid cell qualifies as a cyclone center when its pressure $p_c$ lies below that of all eight neighbors and the surrounding field keeps a sufficient inward gradient. The gradient test compares $p_c$ with the mean pressure $\overline{p}(d)$ on a one-cell ring at distance $d$ from the candidate,

$$\frac{\overline{p}(d) - p_c}{d} \;\geq\; \gamma, \qquad d = 1000 \text{ km}, \quad \gamma = 750 \text{ Pa}\,(1000 \text{ km})^{-1}, \tag{1}$$

and candidates over terrain above 1500 m are discarded. Centers closer than 1000 km are treated as one multicenter system. The cyclone area $A$ is obtained by growing closed isobars around each center at a 200-Pa interval until a contour fails to close, and the last closed isobar defines the edge pressure $p_e$ (Fig. 2b). Each cyclone carries the depth and the area-equivalent radius

$$D = p_e - p_c, \qquad R = \sqrt{A/\pi}. \tag{2}$$

Tracking links the centers of consecutive six-hourly fields (Fig. 2c). For every active cyclone a first-guess position is projected from its previous displacement, damped to three quarters,

$$\mathbf{x}_q = \mathbf{x}_t + 0.75\,(\mathbf{x}_t - \mathbf{x}_{t-1}), \tag{3}$$

and a center of the next field qualifies as the continuation when it lies within $v_{\max}\Delta t$ of both the current position $\mathbf{x}_t$ and the projection $\mathbf{x}_q$, with $v_{\max} = 150$ km h$^{-1}$ and $\Delta t = 6$ h. Among the qualifying candidates the one nearest to the projection continues the track. Centers left unmatched at the new time start tracks as genesis events, and tracks left unmatched end as lysis. Only systems poleward of 30° latitude enter the analysis.

Several track-based diagnostics are derived. Track density counts every six-hourly cyclone position, system density counts each cyclone once at its mean position, and genesis density counts the first point of each track. Tracks that merely cross the boundaries of the monthly processing windows would appear as spurious genesis or lysis events, and such truncated events (about 7% of the total) are removed. For every cyclone the lifespan, the maximum depth relative to the outermost closed isobar, the minimum central pressure, the mean cyclone area, and the propagation speed are recorded.

## 2.3 Gridded storm coverage

A complementary Eulerian diagnostic measures how long each location spends under cyclone influence. At every six-hourly time step a binary mask flags the grid cells that lie inside a square search window of half-width $\sqrt{A}$ grid cells around each cyclone center and whose pressure lies below the edge pressure of that cyclone,

$$M(\mathbf{x}, t) = 1 \quad \text{where } p(\mathbf{x}, t) < p_e \text{ inside the window}, \tag{4}$$

and the masks are regridded to T63 and accumulated into monthly storm days, one quarter of the flagged six-hourly steps. Storm days at a point therefore scale with the product of cyclone number and cyclone area footprint. The two factors can carry opposite seasonal cycles, and over the Southern Ocean the larger size of summer cyclones raises summertime storm days even though cyclone counts peak in winter. Frequency statements in this paper consequently rest on the track-based densities, while storm days serve as a smooth field for maps and for measures of cyclone influence time. The gridded product covers all twelve phases. An early processing inconsistency in the $\omega = 30°$ mask product was traced to a stale intermediate file and removed by regenerating the storm masks of all twelve experiments with one pipeline version, and the regenerated fields were verified month by month against the Lagrangian statistics.

An Eulerian measure of synoptic variability provides a further check that involves no tracking at all. Bandpass sea level pressure variability is computed from the six-hourly fields with the 24-h difference filter of Wallace et al. [1988], which isolates fluctuations on the synoptic time scales that define the classical storm tracks [Blackmon, 1976], and its monthly standard deviation is analyzed with the same basins, seasons, and response measures (appendix Fig. A1).

## 2.4 Baroclinicity diagnostics

The maximum Eady growth rate serves as the measure of baroclinicity [Eady, 1949, Lindzen and Farrell, 1980, Hoskins and Valdes, 1990]. It is computed as $\sigma = 0.31\,|f|\,|\partial U/\partial z|\,N^{-1}$ in the 925–700 hPa layer from monthly wind, temperature, and geopotential height. The response to precession is quantified between the end members p090 and p270, which place perihelion at the June and December solstices. Because the difference fields are dipolar, plain box averages cancel, and the response amplitude is instead defined as the root-mean-square of the difference over a basin, normalized by the climatological growth rate of the season. A substitution decomposition in the style of Camargo et al. [2007] splits the difference field into a vertical-shear factor and a static-stability factor, and the share of each factor is obtained by projecting it onto the full difference pattern. The basin averages use the storm-track boxes defined below. This choice matters in the Southern Hemisphere, where a sector beginning at 30°S would admit the seasonal signal of the subtropical winter jet and disguise it as a storm-belt response.

## 2.5 Basins and statistics

Three storm-track basins are analyzed, the North Atlantic (30–75°N, 80°W–20°E), the North Pacific (30–75°N, 120°E–120°W), and the Southern Ocean (40–75°S, all longitudes). Basin means are area weighted. The size of the precessional response is reported as the range across the twelve phases, the maximum minus the minimum of the 30-year mean, expressed as a percentage of the phase mean. Uncertainty estimates use the interannual spread of the 30 individual years in each phase, and the significance of the end-member difference maps is assessed with a two-sided Welch $t$ test at each grid point. The local-summer contrasts between extreme phases far exceed the interannual noise, with $p$ values below $10^{-12}$ in all three basins.

# 3 Results

## 3.1 Winter storm tracks and what storm days measure

The three storm tracks of the ensemble live in winter, exactly as observed. Seasonal-mean genesis counts peak in DJF over the North Atlantic and the North Pacific and in JJA over the Southern Ocean, and system counts follow the same calendar in every phase. Cyclones also cut deepest in the cold season, with winter maximum depths exceeding their summer values by roughly half in the northern basins. The winter identity of the storm season is therefore common to both hemispheres and to all twelve orbital configurations (Fig. 4a–c; Table 1).

The gridded storm-day field tells a partly different story, and the difference is instructive. Basin-mean storm days over the Southern Ocean stay high through austral summer although summer cyclones are fewer. The elevation reflects cyclone size rather than cyclone number. Summer systems there are about twenty percent larger in area than their winter counterparts, and storm days count the time a grid cell spends inside any cyclone, so the product of number and footprint hides the winter peak in counts. A diagnostic of cyclone influence time is not a diagnostic of cyclone frequency, and the two must be read together. Frequency statements below therefore rest on the track-based densities, while storm days provide the smooth fields used for maps.

## 3.2 The summer paradox

The annual-mean view sets the baseline (Fig. 3). The twelve annual climatologies are close to interchangeable, and the anomalies of each phase from their common mean form weak, smoothly rotating dipoles along the storm tracks. The basin series beneath the maps quantify how little survives in the annual mean, with across-phase ranges of 10%, 17% and 4% in the North Atlantic, the North Pacific and the Southern Ocean, varying smoothly with $\omega$ and already hinting at the interhemispheric antiphase, since the northern basins reach their minima near $\omega = 90°$ where the Southern Ocean peaks.

Precession barely touches the annual mean but reorganizes the seasons, and the season it selects is the quiet one. The seasonal cycles of basin-mean track density keep their winter peak in every phase and fan out across the phases almost exclusively in local summer (Fig. 4a–c). The across-phase range makes this quantitative (Fig. 4d–f). In the North Pacific the annual range is 7% of the phase mean while the JJA range reaches 44%. The North Atlantic behaves the same way with 7% against 25%. The Southern Ocean repeats the pattern half a year out of phase, with an annual range of 4% against 22% in DJF. System and genesis densities confirm the calendar in every basin, and the influence-time metric swings harder still, with a storm-day range of 68% in North Pacific summer (Table 1). The sensitive season is local summer everywhere, and the small annual signal is what survives of a substantial summer swing after the stable winter is averaged in.

A tracking-free view points the same way (appendix Fig. A1). Bandpass sea level pressure variability responds most strongly in JJA over the North Pacific, at 36% of its climatology against 9% in winter, and winter holds the weakest response of all seasons in both northern basins. This amplitude-weighted metric shifts part of the response within the warm half-year. Over the North Atlantic it peaks in early autumn, matching the secondary September lobe of the forcing itself (Fig. 1c), and over the Southern Ocean its apparent winter share lies on the equatorward flank of the basin box, while poleward of 50°S austral summer leads. The warm-season concentration and the winter quiescence are therefore not artifacts of the tracking algorithm.

## 3.3 Orbital anchoring and the interhemispheric mirror

The summer response is anchored to the orbit, not to the calendar. When the basin-mean track-density anomaly is drawn as a function of calendar month and precession phase, the anomaly ridge climbs through the diagram along the calendar month of perihelion (Fig. 4g–i). The sign is fixed by the orbital geometry. Summers spent near perihelion are quiet and summers spent near aphelion are stormy. North Pacific summer activity reaches its minimum at $\omega = 90°$, the configuration with perihelion at the June solstice, where it falls 31% below the phase mean, and its maximum of +37% near $\omega = 270°$ (Fig. 7a). The North Atlantic swings between −16% and +15% at the same phases. The Southern Ocean mirrors this behavior, with a DJF maximum at $\omega = 60°$ and a minimum at $\omega = 210°$, because a perihelion that falls in northern summer is an aphelion summer for the south. One lever thus drives the two hemispheres in antiphase across the precession cycle. The extreme-phase contrasts far exceed interannual variability, with Welch $p$ values below $10^{-12}$ in all three basins.

## 3.4 The dynamical bridge runs through summer shear

The seasonal selectivity begins in the forcing itself. Averaged over the northern storm-track band from 40 to 70°N, the seasonal forcing amplitude peaks at 105 W m$^{-2}$ in JJA and collapses to 14 in DJF, and the mirror band in the south peaks at 119 W m$^{-2}$ in DJF against 47 in JJA (Fig. 1). Perihelion can only amplify a season that already receives sunlight, so the orbital lever presses on local summer in each hemisphere, keeps a substantial autumn shoulder, and leaves deep winter almost untouched.

Baroclinicity translates the pressure on the seasons into the storm response. The Eady growth rate difference between the end members p090 and p270 is largest in local summer within every storm basin (Fig. 5). Relative to the seasonal climatology the summer response reaches 31% in the North Atlantic, a striking 92% in the North Pacific, and 21% in the Southern Ocean, in each case the seasonal maximum for that basin. A substitution decomposition attributes the pattern almost entirely to the vertical-shear factor, whose projection accounts for 103–109% of the difference field in the three basins, while the static-stability factor contributes near zero. The chain is short and uniform. Perihelion summers warm the summer hemisphere, flatten the meridional temperature gradient, weaken the thermal-wind shear, and starve the baroclinic growth on which summer cyclones depend. The storm anomalies appear somewhat poleward of the growth-rate anomalies, consistent with the downstream development of baroclinic waves [Chang et al., 2002], and this offset is why pattern measures rather than same-latitude box averages are used throughout.

Winter tells a complementary story. The winter growth rate still shifts by 16–18% between the end members, yet the winter storm response stays below 12% in every basin, a point taken up in the discussion.

## 3.5 Spatial expression, and what precession actually moves

The end-member difference maps show where the summer signal lives (Fig. 6). In the annual mean the p090 minus p270 difference is weak and patchy. In JJA a pronounced dipole spans the northern basins, with fewer storm days through the midlatitude core and more along the Arctic flank at perihelion summer, the signature of a poleward shift joined to an overall summer weakening. In DJF the significant signal migrates to the Southern Ocean and forms a circumpolar banded structure of opposite sign. About half of the JJA grid points in the northern extratropics pass the 95% significance test, against a far smaller fraction in the annual mean.

The track displacement behind these dipoles can be made explicit (appendix Fig. A2). The storm-day-weighted latitude of the North Pacific summer track migrates from 53°N when perihelion falls in northern winter to 61°N when it falls in northern summer, a poleward excursion of 8° that peaks exactly at $\omega = 90°$, and the track-density-weighted latitude follows the same curve with a 5° swing. The North Atlantic summer track moves 4.4° in the same direction, and the Southern Ocean summer track mirrors the rule with a poleward excursion of 3.8° when perihelion approaches austral summer. A perihelion summer therefore thins the storm population and displaces the survivors poleward, in step with the poleward shift of the baroclinic zone seen in the growth-rate dipole of Fig. 5a. The winter track latitude stays within 1.5–2.8° across the entire cycle, and its small extremes do not align with the solstitial phases, which extends the winter invariance from storm counts to storm position. The shoulder seasons fall in between and carry a telling phase structure, because the autumn excursion peaks near $\omega = 150°$, the configuration that places perihelion in late summer, so the poleward displacement follows the perihelion month through the calendar exactly as the occurrence anomaly does in Fig. 4g–i. The annual-mean track latitude varies by no more than 1.3° in any basin.

The orbital anchoring of occurrence and position condenses into a single statement (Fig. 8). A first-harmonic fit across the phases gives, for every calendar month, the value of $\omega$ at which storm activity reaches its minimum and the track its poleward maximum, and through the sunlit half-year both sequences climb in parallel with the line that places perihelion in that month. The response lags the perihelion calendar by a basin-dependent interval, near one month in the North Pacific, one to two months in the North Atlantic, and slightly over two months in the Southern Ocean, an ordering consistent with the growing oceanic character of the three sectors and with surface thermal inertia setting the delay. Winter months fall off the line, where the response amplitude is too small to anchor. The storm tracks are phase locked to the perihelion calendar wherever the lever can reach them.

What precession moves is above all the number and location of cyclones, not their translation speed, and only regionally their depth (Fig. 7b). Local-summer ranges of the frequency and position metrics span 16–68% across the basins. Propagation speed stays within 2–5% everywhere. Cyclone depth separates the basins, and the diagnostics are deliberately kept apart in the manner of Zappa et al. [2013]. Over the Southern Ocean the maximum depth varies by only 4% across the phases while frequency varies by 22%, a clean decoupling of frequency from intensity. Over the North Pacific the summer depth range reaches 33% and the lifespan range 23%, so there the whole cyclone population weakens together when perihelion falls in summer, fewer systems that are also shallower and shorter lived. The North Atlantic sits between these regimes. The common element across basins is that occurrence responds most, while the intensity response ranges from absent to substantial depending on how strongly the summer baroclinic reservoir is drained.

## 3.6 What the rain records

The hydrological expression of the cycle runs opposite to the dynamical one (appendix Fig. A3). Detected cyclones deliver about one third of extratropical precipitation in the ensemble, in line with reanalysis-based attribution [Hawcroft et al., 2012]. In North Pacific summer the precipitation falling inside each storm strengthens by up to 15% exactly where storm counts collapse, and the two series are nearly perfectly anticorrelated across the phases ($r = -0.95$). A perihelion summer holds a warmer and moister atmosphere, so the surviving storms rain harder, but the loss of storms wins, and total summer precipitation still falls by 17% between the end members. The share of precipitation delivered by storms falls with it, by 9.5 percentage points across the cycle in North Pacific summer against about 3 in the annual mean. Precession therefore forces extratropical hydroclimate in two opposing ways at once, fewer storms against wetter storms, and the two must be kept apart in interpretation just as counts and depths must.

# 4 Discussion

## 4.1 Why the lever lands in summer

The seasonal selectivity of the storm response is set by orbital geometry before any model physics enters. Precession redistributes insolation in proportion to the insolation a season already has, so the winter hemisphere offers the lever almost nothing to press on [Berger, 1978]. The monthly forcing amplitude in Fig. 1 makes the floor explicit, with midwinter values near 20 W m$^{-2}$ against summer values above 125 W m$^{-2}$ at the latitudes of the storm tracks. This part of the result is model independent and would reappear in any climate model, because it restates the geometry that also underlies the summer-energy view of orbital forcing [Huybers, 2006]. The novelty is not that summer receives the forcing, which the monsoon literature established long ago [Kutzbach and Guetter, 1986, Bosmans et al., 2015], but that the summer forcing finds a winter phenomenon and reorganizes it through its quiet season.

## 4.2 Cool summers are stormy summers

The mechanism that converts summer insolation into summer storms is the thermal wind. Perihelion summers warm the summer hemisphere most strongly over land and at high latitudes, flatten the meridional temperature gradient, and weaken the vertical shear that feeds baroclinic growth. The factor decomposition confirms that the shear term carries essentially the whole Eady growth rate response in all three basins, while static stability contributes almost nothing. The same chain, with greenhouse warming in place of perihelion, is now observed in the present-day Northern Hemisphere, where amplified summer warming at high latitudes has weakened the summer circulation, the eddy kinetic energy, and the number of strong summer cyclones [Coumou et al., 2015, Chang et al., 2016, Gertler and O'Gorman, 2019]. Projections extend the analogy, with CMIP-class models thinning summer cyclone activity and the associated baroclinicity through the same shear pathway [Lehmann et al., 2014, Priestley and Catto, 2022]. The correspondence should be read as a shared mechanism rather than a shared scenario, because the warming trend alters the two hemispheres together whereas precession swings them in antiphase. Under both forcings the rule is the same, and it is the rule in the title. Cool summers are stormy summers.

### 4.3 The orbital invariance of the winter storm track

The stability of winter deserves as much attention as the mobility of summer. Winter storm activity varies by less than about a tenth across the entire precession cycle in every basin, and this is only partly a matter of weak winter forcing. The winter Eady growth rate still shifts by 16–18% between the end members, yet winter cyclone numbers barely respond. The winter storm track appears saturated in the sense that ample baroclinicity is available in every phase, so that moderate changes in the growth rate do not translate into changes in occurrence. Whatever its ultimate cause, the practical consequence is notable. The delivery of winter precipitation to the midlatitude and subpolar continents, the season and pathway that build snowpack and feed ice sheets, is essentially fixed across orbital configurations of the precession cycle. A transient simulation of the last millennium likewise found cyclone counts nearly insensitive to external forcing [Raible et al., 2018, Hess and Chemke, 2025], and the present result extends that insensitivity to the far larger insolation excursions of the orbital band, for winter alone.

### 4.4 One lever, three basins, and what the records would see

The three basins express one mechanism with different accents, and the diagnostics must be kept apart to see it. Occurrence responds everywhere, with local-summer ranges of 16–68% in frequency and position metrics, while propagation speed is inert. Intensity separates the basins. The Southern Ocean decouples cleanly, with a 4% depth range against a 22% frequency range, the North Pacific weakens as a whole population in perihelion summers, and the North Atlantic lies between. The pattern echoes the long-standing conclusion from anthropogenic scenarios that cyclone number and cyclone intensity respond to forcing through different processes and need not move together [Bengtsson et al., 2009, Ulbrich et al., 2009, Catto et al., 2019]. The water cycle sharpens the contrast, because per-storm precipitation strengthens in the very summers that lose storms, an orbital counterpart of the warming-scenario finding that hydrological responses outrun dynamical ones [Bengtsson et al., 2009]. For the record this matters twice over. Extratropical precipitation is delivered disproportionately by cyclones and their fronts [Hawcroft et al., 2012, Catto and Pfahl, 2013], and the storm-attributed share of precipitation swings by up to ten percentage points across the cycle in North Pacific summer, so a precipitation archive senses the competition between fewer storms and wetter storms rather than either alone. The storm-day diagnostic adds its own caution, because cyclone influence time folds cyclone size into cyclone number, and over the Southern Ocean the seasonal cycle of size alone can invert the apparent storm season. Any proxy that integrates storm influence rather than storm counts inherits this ambiguity.

For the paleoclimate record the seasonal anatomy carries a concrete and testable message. An archive with summer-biased sensitivity in the midlatitude storm belts should carry a pronounced precession signal in storminess, while an annually integrating archive should carry almost none. The seasonal bias of an archive can therefore fabricate or erase an orbital storm signal exactly as it does for temperature [Laepple et al., 2011]. The inter-hemispheric mirror sharpens the prediction, because summer-sensitive storminess records from the two hemispheres should vary in antiphase at the precession period, the extratropical counterpart of the monsoon seesaw [Wang et al., 2008]. Existing archives make the test realistic. Precession-band signals are already established in the northern midlatitude westerlies [Li et al., 2019] and in the composition of North Pacific dust, where the precession imprint sits in the moisture-sensitive component while the annually integrating flux follows obliquity [Zhong et al., 2024], and a winter-integrating dust-strength record shows correspondingly weak precession power [Chen et al., 2014]. In the south the westerly belt carries strong 19 and 23 kyr cycles at its subtropical edge [Lamy et al., 2019], with the orbital band recorded at a site depending on its latitude within the belt [Kaiser et al., 2024]. The subtropical location of that southern signal is consistent with the present results, because the winter half of the precessional baroclinicity response resides in the subtropical jet equatorward of the storm belt, while the storm-belt response is a summer phenomenon. Records of the southern westerlies also show orbital-band migrations of the belt itself [Lamy et al., 2010, Varma et al., 2012], and glacial-state cyclone modeling has shown how strongly reconstructed storminess depends on which cyclone property a proxy senses [Pinto and Ludwig, 2020]. A paired-hemisphere test at the precession band is within reach of existing archives.

## 4.5 Limitations

Several limitations bound these conclusions. The results rest on one model, one tracking algorithm, and an idealized orbital configuration with eccentricity fixed near its Quaternary maximum, so the simulated amplitudes represent an upper bound that scales with $e \sin \omega$ rather than a reconstruction of any particular epoch. Obliquity is fixed at 24.5°, and interactions between precession and obliquity or ice volume are outside the design. The simulations are equilibrium slices, so transient effects along the precession cycle are not represented. The T63 atmosphere resolves the storm tracks but not mesoscale substructure, and low-level Eady growth rates in the Southern Hemisphere carry a documented sensitivity to boundary-layer stability [Simmonds and Lim, 2009]. The gridded storm-day product required one uniform regeneration of the storm masks before all twelve phases could be used together, a reminder that multi-batch processing of tracked-cyclone products deserves explicit version control.

# 5 Conclusions

One rule spans the three storm tracks of this precession ensemble. The orbital lever reaches extratropical cyclones through local summer, the quiet season, while the winter storm track stays essentially fixed across the entire cycle. Summers spent near perihelion are calm and summers near aphelion are stormy, and because one hemisphere's perihelion summer is the other's aphelion summer, midlatitude storminess in the two hemispheres swings in antiphase at the precession period. The signal travels through shear-driven summer baroclinicity and expresses itself chiefly in how many cyclones occur and where they run, with per-cyclone intensity responding only where the summer baroclinic reservoir is drained most deeply. Both expressions are phase locked to the calendar month of perihelion, with a delay of one to two months that grows with the oceanic character of the basin. For the proxy record the lesson is direct. Orbital storminess signals should be sought in summer-sensitive midlatitude archives, and they should appear with opposite sign in the two hemispheres.

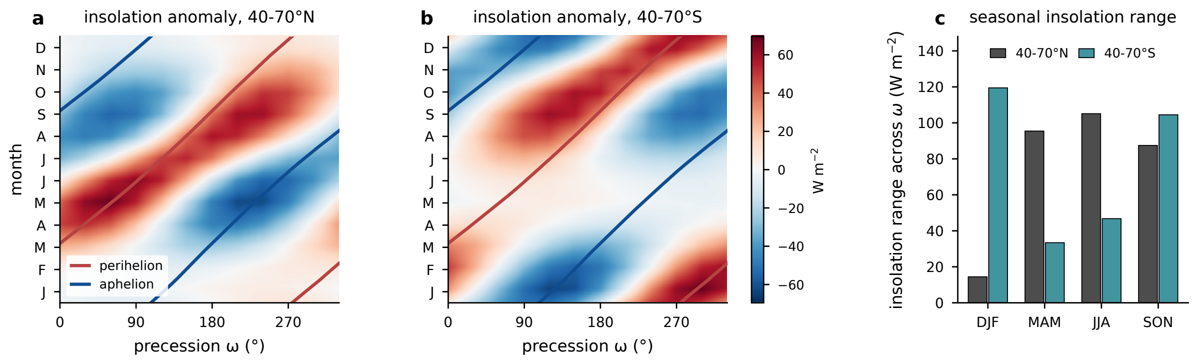


Figure 1: Precessional insolation forcing from orbital geometry with eccentricity 0.058 and obliquity 24.5°. (a) Monthly insolation anomaly relative to the phase mean, averaged over the storm-track band 40–70°N, as a function of calendar month and precession phase $\omega$, with the perihelion (red) and aphelion (blue) calendar months of each phase overlaid. (b) As in (a), but for 40–70°S. (c) Seasonal forcing amplitude in the two bands, the seasonal mean of the monthly maximum minus minimum insolation across the twelve phases. Units: W $m^{-2}$.

Table 1: Three-basin comparison of the precessional storm response. Ranges are max minus min across the twelve phases as a percentage of the phase mean. Storm and sensitive seasons refer to local winter and local summer. The storm season is diagnosed from cyclogenesis counts, the sensitive season from the seasonal storm-day range. The EGR response is the RMS p090−p270 difference over the basin sector relative to its climatology, and the shear share is the projection of the vertical-shear factor onto the response pattern.

| Basin | Storm season | Sensitive season | Storm-day range ann/summer (%) | Summer max/min $\omega$ (°) | EGR response summer (%) | Shear share (%) |
|---|---|---|---|---|---|---|
| North Atlantic | DJF | JJA | 10 / 32 | 240 / 90 | 31 | 107 |
| North Pacific | DJF | JJA | 17 / 68 | 270 / 90 | 92 | 106 |
| Southern Ocean | JJA | DJF | 4 / 16 | 60 / 210 | 21 | 109 |

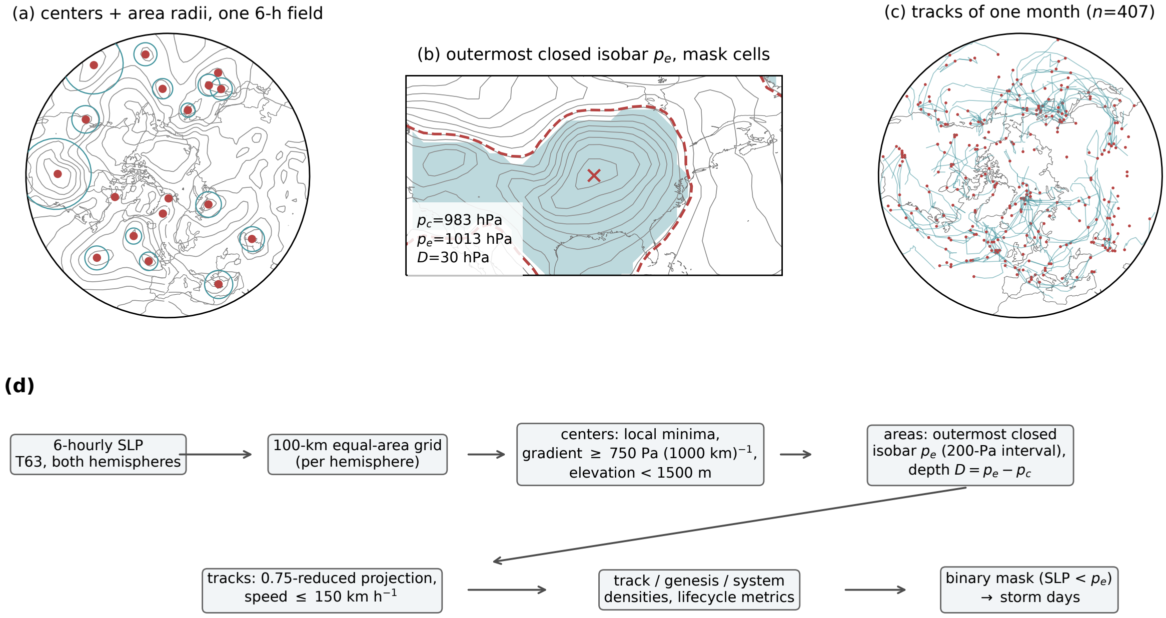


Figure 2: Cyclone detection and tracking procedure. (a) One six-hourly sea level pressure field of phase p000 (contours, 8-hPa interval) with all detected cyclone centers poleward of 30°N (dots) and their area-equivalent radii $R = \sqrt{A/\pi}$ (circles), 15 January of one model year. (b) The deepest system of that field, with the outermost closed isobar $p_e$ (dashed), the cyclone center (cross), the central and edge pressures and depth annotated, and the binary mask cells (shading) defined by SLP $< p_e$ inside the $\sqrt{A}$ search window; contour interval 4 hPa. (c) All cyclone tracks of the same month (lines) with genesis points (dots). (d) Processing chain with the key thresholds of Eqs. (1)–(4).

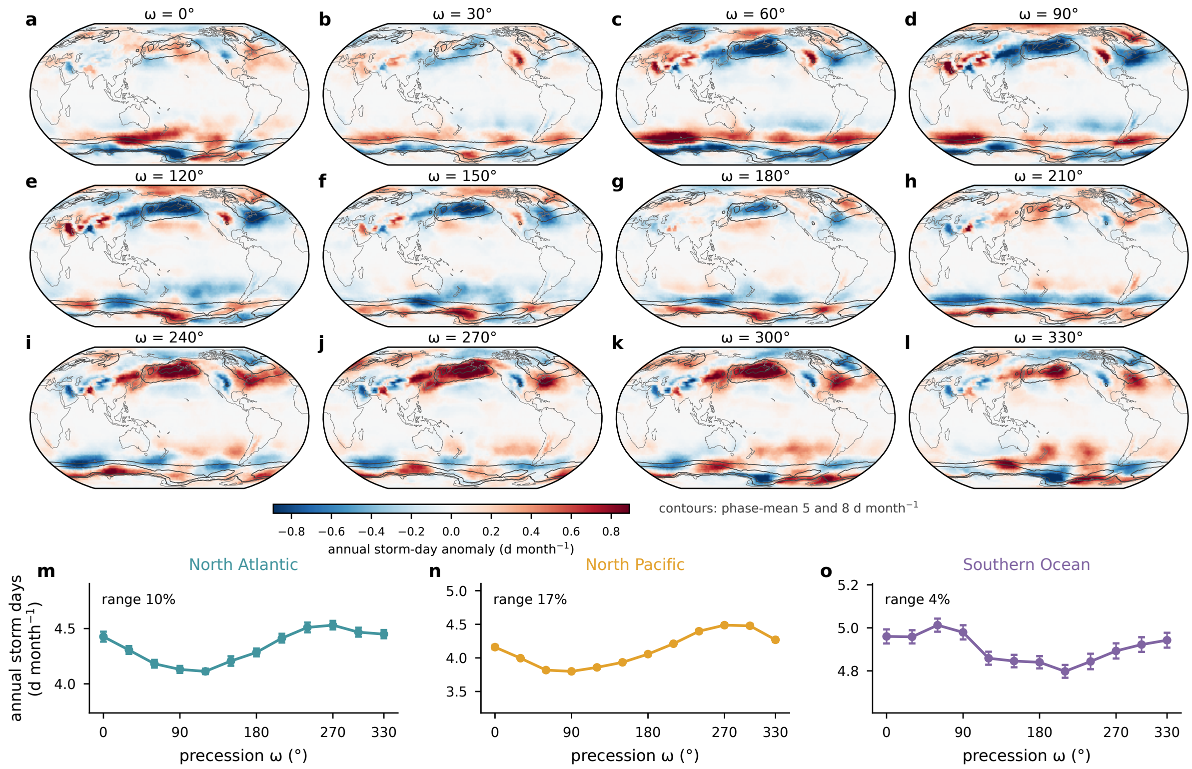


Figure 3: Annual-mean storm days across the precession cycle. (a)–(l) Annual storm-day anomaly of each phase relative to the twelve-phase mean (shading), with the phase-mean climatology outlined at 5 and 8 d month$^{-1}$ (contours). (m)–(o) Basin-mean annual storm days against $\omega$ with interannual standard errors; the across-phase range as a percentage of the phase mean is annotated in each panel.

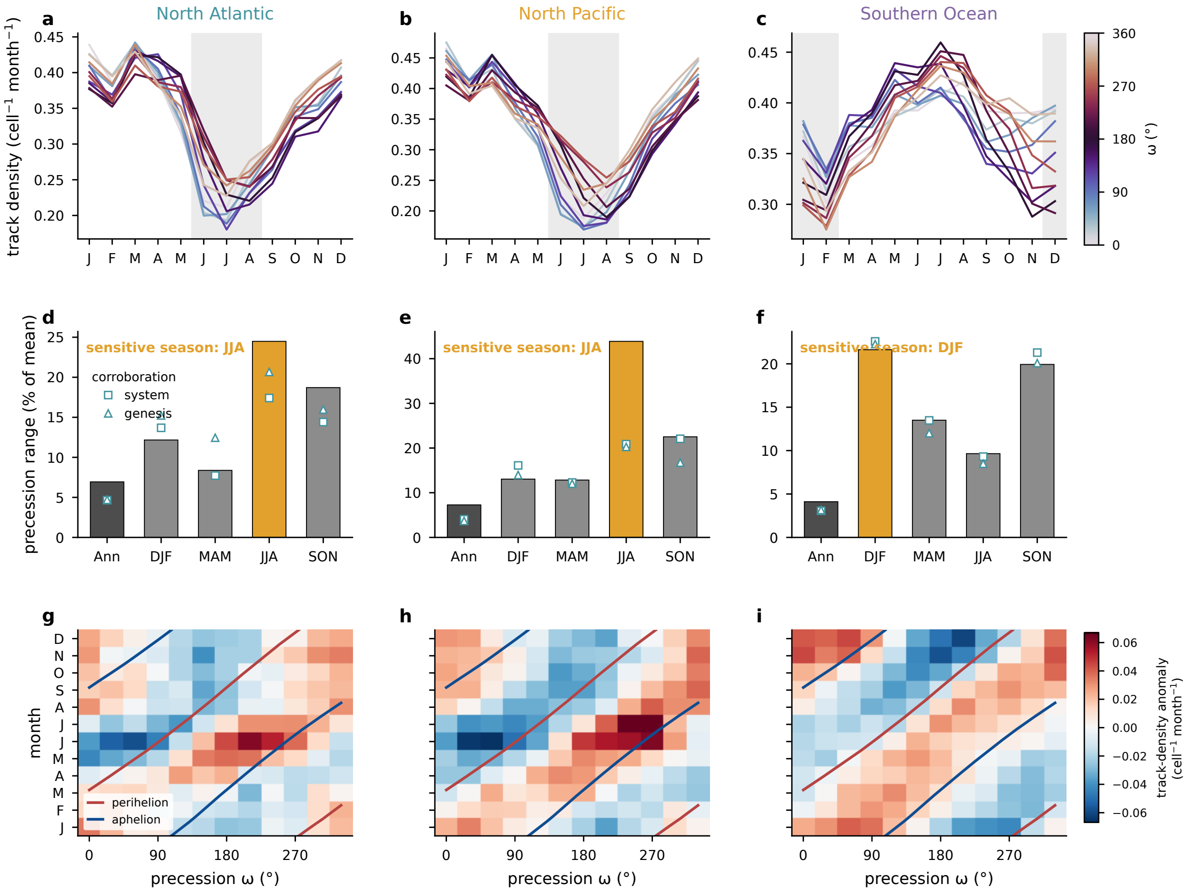


Figure 4: Seasonal concentration of the precessional storm response, from the Lagrangian tracks. (a)–(c) Basin-mean track density (cell$^{-1}$ month$^{-1}$) by calendar month for the North Atlantic, the North Pacific, and the Southern Ocean, one line per phase colored by $\omega$, with local summer shaded. (d)–(f) Across-phase range of track density (maximum minus minimum, percent of the phase mean) for the annual mean and the four seasons, with the local-summer bar highlighted; open symbols give the corresponding ranges of system and genesis densities. (g)–(i) Basin-mean track-density anomaly relative to the phase mean as a function of calendar month and precession phase (unsmoothed), with the perihelion (red) and aphelion (blue) calendar months of each phase overlaid.

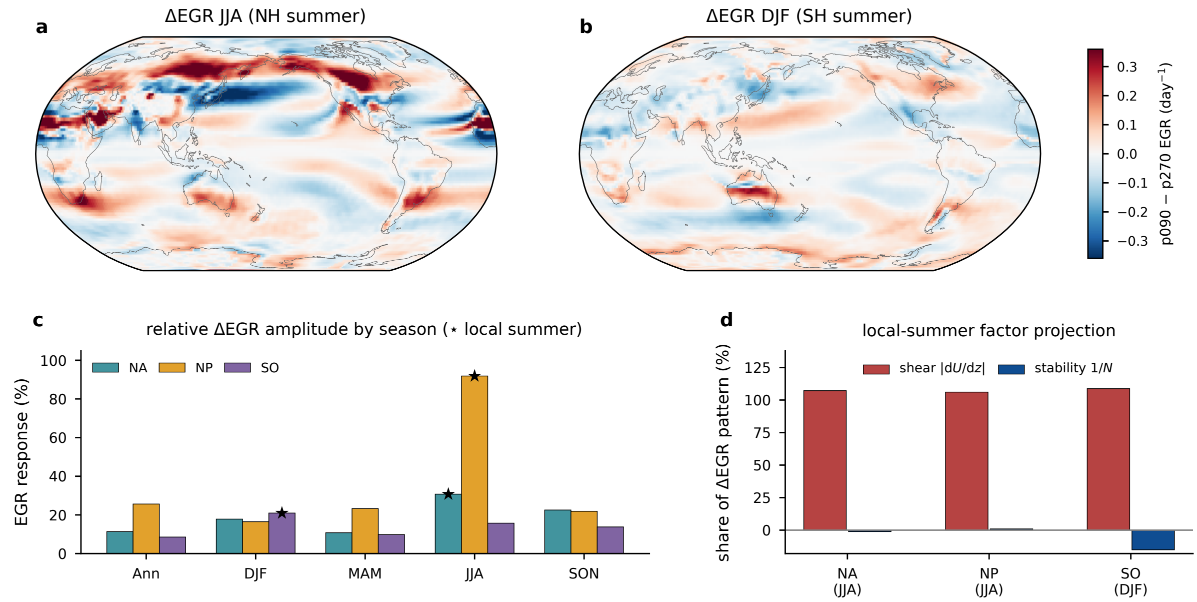


Figure 5: Precessional response of the maximum Eady growth rate (925–700 hPa). (a) p090 minus p270 difference for JJA and (b) for DJF (day$^{-1}$). (c) Seasonal response amplitude in each basin, the RMS of the p090 minus p270 difference over the storm box as a percentage of the seasonal climatology, with stars marking local summer. (d) Projection shares of the vertical-shear and static-stability factors in the local-summer difference pattern.

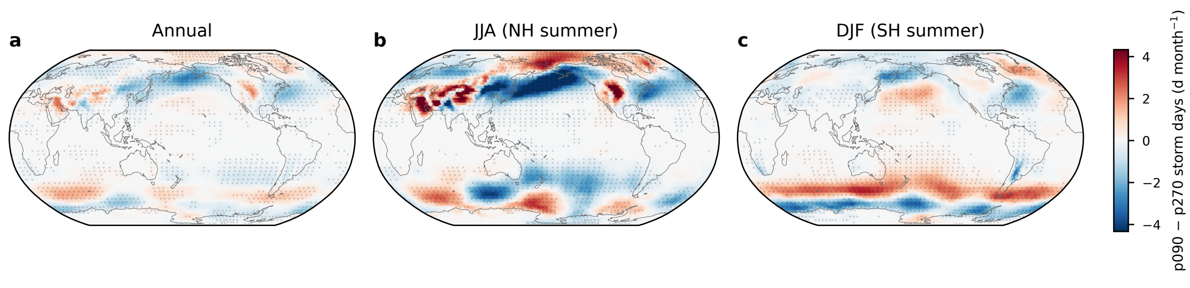


Figure 6: Difference in storm days (d month$^{-1}$) between p090 and p270 for (a) the annual mean, (b) JJA, and (c) DJF, on a common color scale. Stippling marks grid points where the difference is significant at the 95% level based on a two-sided Welch $t$ test over the 30 simulated years.

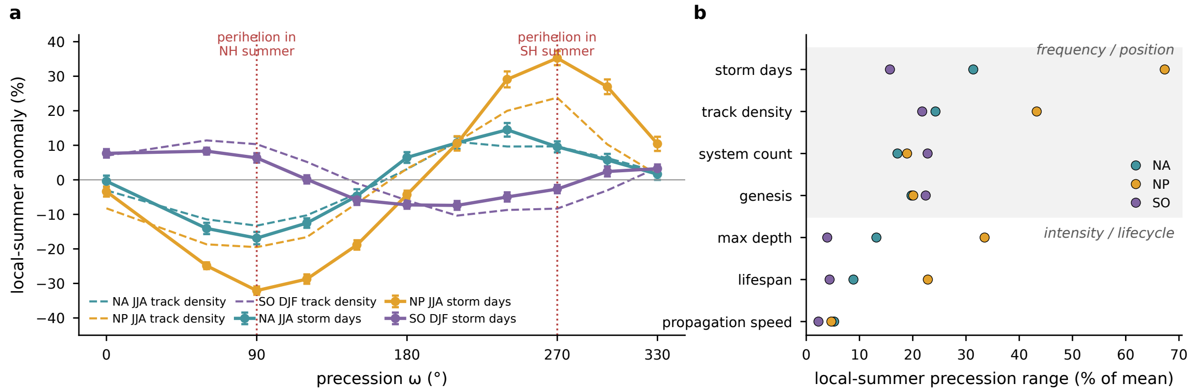


Figure 7: (a) Local-summer storm activity across the precession cycle, shown as the percent anomaly of basin-mean storm days from the phase mean (solid, with interannual standard errors) and of track density (dashed), for the North Atlantic and the North Pacific in JJA and for the Southern Ocean in DJF; dotted vertical lines mark the phases with perihelion in NH and SH summer. (b) Local-summer across-phase range for frequency and position metrics and for intensity and lifecycle metrics in each basin.

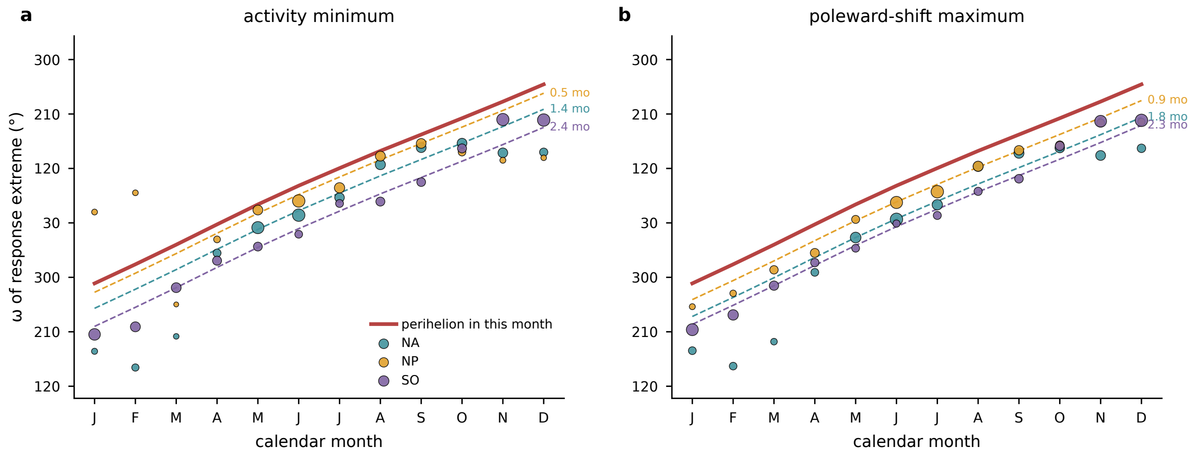


Figure 8: Phase locking of the storm response to the perihelion calendar. For each calendar month and basin, the phase of the first harmonic across the twelve precession phases of (a) basin-mean storm days, shown as the $\omega$ of the activity minimum, and (b) the storm-day-weighted track latitude, shown as the $\omega$ of the poleward maximum. Marker size scales with the harmonic amplitude. The red line gives the $\omega$ that places perihelion in that calendar month, and dashed lines repeat it delayed by each basin's median lag over the amplitude-strong months (annotated in months).

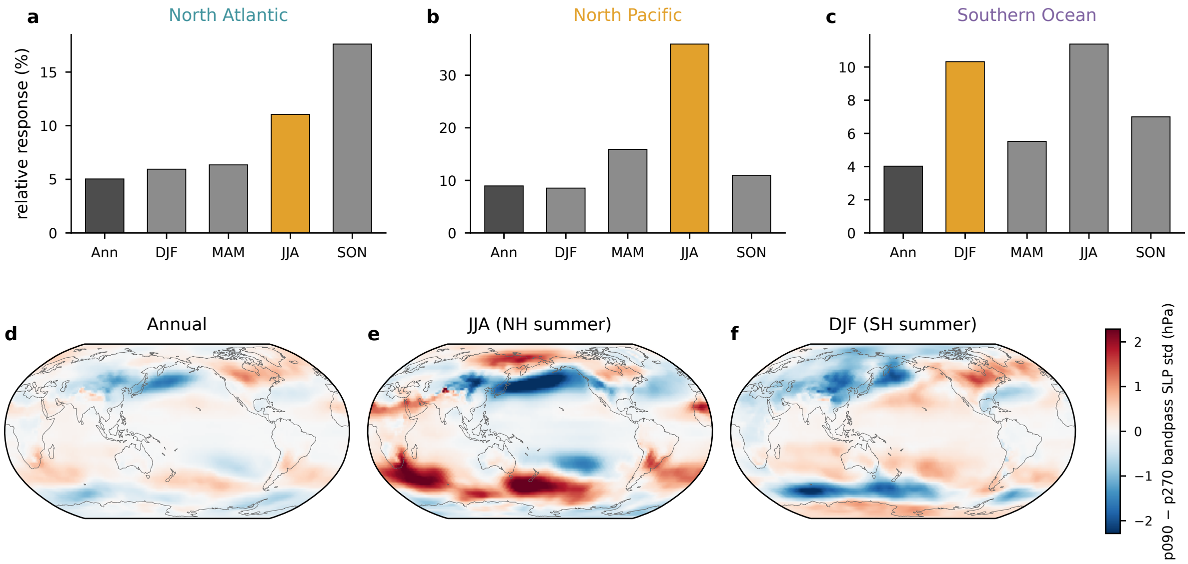


Figure A1: Tracking-free Eulerian storm-track metric, the monthly standard deviation of 24-h differences of six-hourly sea level pressure. (a)–(c) Relative seasonal response in each basin, the RMS of the p090 minus p270 difference over the storm box as a percentage of the seasonal climatology (as in Fig. 5c), with the local-summer bar highlighted. (d)–(f) The p090 minus p270 difference of the same quantity for the annual mean, JJA, and DJF (hPa).

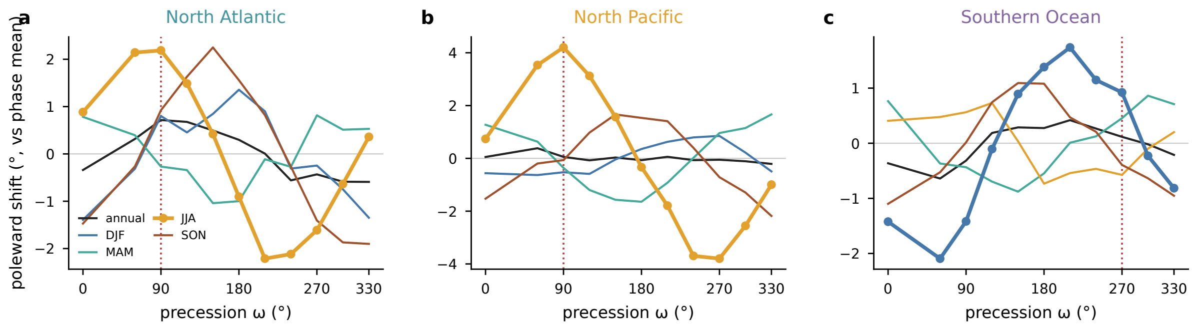


Figure A2: Storm-track latitude across the precession cycle, the storm-day-weighted mean latitude over each basin sector shown as the poleward anomaly from each season's phase mean, for (a) the North Atlantic, (b) the North Pacific, and (c) the Southern Ocean. Lines give the annual mean and the four seasons, with the local summer of each basin drawn heavy with markers. Vertical dotted lines mark the phase with perihelion in the local summer of each basin.

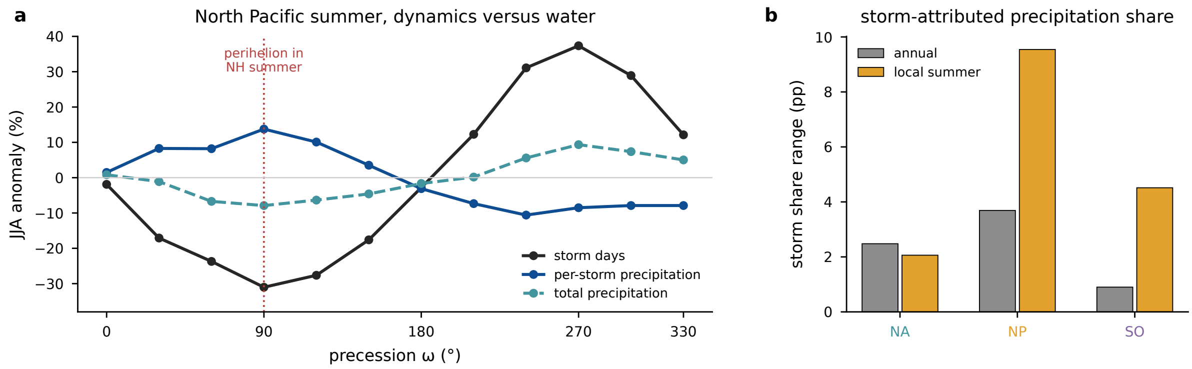


Figure A3: Storm-attributed precipitation across the precession cycle. (a) North Pacific JJA percent anomalies from the phase mean of basin-mean storm days, per-storm precipitation, and total precipitation; the dotted line marks the phase with perihelion in NH summer. (b) Across-phase range of the storm-attributed share of precipitation, accumulated precipitation inside storm footprints divided by total accumulated precipitation (percentage points), for the annual mean and local summer in each basin.

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